

THE CIRCLE IS NOT THE BOUNDARY OF THOMPSON'S GROUP T AND SPACE OF SUBGROUPS OF WREATH PRODUCTS

MARTÍN GILABERT AND EDUARDO SILVA

ABSTRACT.

1. THOMPSON'S GROUP T

Thompson's group T acts on the circle S^1 , and for any probability measure μ on T there is a unique μ -stationary measure on S^1 . **are there any assumptions on the measure? non-degenerate probably? some moment assumption?** The objective of this section is to prove that (S^1, ν) is not the Poisson boundary of the μ -random walk on T . This answers a question asked in the second item of Section 6 of [Der13], and which is asked again in [Nav18, Question 19] for finitely supported, non-degenerate and symmetric probability measures on T .

Theorem 1.1. *Let μ be a non-degenerate finitely supported probability measure on Thompson's group T . Let ν be the unique stationary measure on the circle S^1 such that (S^1, ν) is a μ -boundary of T . Then (S^1, ν) is not the Poisson boundary of (T, μ) .*

2. THE CONDITIONAL ENTROPY CRITERION AND NON-MAXIMALITY OF A μ -BOUNDARY

The following criterion goes back to [Kai85, Theorem 2] (see also [Kai00, Theorem 4.6])

Theorem 2.1 (Kaimanovich's conditional entropy criterion). *Let G be a countable group and let μ be a probability measure on G with $H(\mu) < \infty$. Consider a μ -boundary (X, λ) of G . Then (X, λ) is the Poisson boundary of (G, μ) if and only if for λ -almost every $\xi \in X$ we have $h^\xi(\mu) = 0$.*

2.1. Conditional version of Anna's entropy method.

Definition 2.2. Let G be a countable group and consider a probability measure μ on G with $H(\mu) < \infty$. Consider a non-trivial μ -boundary (X, λ) of G . Denote by $\pi : G^{\mathbb{Z}^+} \rightarrow X$ the associated boundary map.

Consider a measurable subset $Y \subseteq X$ with $\lambda(Y) > 0$. We say that an element $a \in G$ is Y -stable if for every measurable subset $A \subseteq Y$ with $\lambda(A) > 0$ and $\mathbb{P}$ -almost every sample path $\mathbf{w} = \{w_n\}_{n \geq 0} \in G^{\mathbb{Z}^+}$ we have

$$\pi(g\mathbf{w}) \in A \iff \pi(ga\mathbf{w}) \in A,$$

for every $g \in G$ such that $g^{-1}Y \subseteq Y$.

Lemma 2.3. *Let G be a countable group and consider a probability measure μ on G with $H(\mu) < \infty$. Consider a non-trivial μ -boundary (X, λ) of G , and fix a measurable subset $Y \subseteq X$ with $\lambda(Y) > 0$. Let $a \in G$ be a Y -stable element. Then for λ -almost every $\xi \in Y$ and every $g \in G$ such that $g^{-1}Y \subseteq Y$ we have*

$$\frac{d(ga)_*\lambda}{dg_*\lambda}(\xi) = 1.$$

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Proof. Indeed, let $A \subseteq Y$ be any measurable subset with $\lambda(A) > 0$. Then

$$\begin{aligned} g_*\lambda(A) &= \lambda(g^{-1}A) \\ &= \mathbb{P}(\pi(\mathbf{w}) \in g^{-1}A) \\ &= \mathbb{P}(\pi(g\mathbf{w}) \in A) \\ &= \mathbb{P}(\pi(ga\mathbf{w}) \in A) \\ &= \mathbb{P}(\pi(\mathbf{w}) \in (ga)^{-1}A) \\ &= (ga)_*\lambda(A). \end{aligned}$$

□

For λ -almost every $\xi \in X$ one can consider the conditional probability measure $\mathbb{P}^\xi$, so that one has $\mathbb{P} = \int_X \mathbb{P}^\xi d\lambda(\xi)$. These measures are Markov and determine homogeneous (in time, not space) Markov chains on G with transition probabilities

$$\mathbb{P}^\xi(w_{n+1} = g \mid w_n = h) = \mathbb{P}(w_{n+1} = g \mid w_n = h) \frac{dg_*\lambda}{dh_*\lambda}(\xi) = \mu(h^{-1}g) \frac{dg_*\lambda}{dh_*\lambda}(\xi),$$

for each $g, h \in G$. Lemma 2.3 implies that for moments where we multiply by a then we have the value $\mu(a)$ as a transition probability in the conditional random walk. Similarly for the identity element.

Remark 2.4.

Theorem 2.5. *Let $Y \subseteq X$ be a measurable subset with $\lambda(Y) > 0$. Let $a \in G$ be a Y -stable element. Suppose that for λ -almost every $\xi \in Y$ there exist $p, c > 0$ such that for each n there is A_n a set of collections $q = (w, i_1, \dots, i_k)$ such that*

- (1) *For each $q = (w, i_1, \dots, i_k) \in A_n$ we have $k \geq cn$.*
- (2) *this should be written better. The increment g between two "empty spots" should satisfy $g^{-1}Y \subseteq Y$.*
- (3) *For each $q \in A_n$ the set of trajectories $T^{a, e_G}(q)$ is satisfactory.*
- (4) *For each $q_1, q_2 \in A_n$ with $q_1 \neq q_2$ we have $T^{a, e_G}(q_1) \cap T^{a, e_G}(q_2) = \emptyset$.*
- (5) *For each $n \geq 1$ it holds that*

$$\mathbb{P}^\xi \left(\bigcup_{q \in A_n} T^{a, e_G}(q) \right) \geq p.$$

Then $h^\xi(\mu) > 0$ for λ -almost every $\xi \in Y$.

Proof. The proof follows the next steps.

- (1) We have $h^\xi(\mu) = \lim_{n \rightarrow \infty} \frac{H_\xi(\alpha_n)}{n}$, where α_n is the partition where two trajectories $\mathbf{w}, \mathbf{w}'$ are equivalent if and only if $w_n = w'_n$.
- (2) Let ν be the conditional measure of μ on the set $\{e_G, a\}$. Then for every $q \in A_n$ and λ -almost every $\xi \in Y$, the normalized restriction of $\mathbb{P}^\xi$ restricted to $T^{a, e_G}(q)$ is the product measure ν^k . Here is where we use Lemma 2.3.
- (3) The rest of the proof is analogous to Anna's. We have

$$H(\nu^k) = kH(\nu) \geq cnH(\nu).$$

(4) Additionally, we have

$$\begin{aligned} H_\xi(\alpha_n) &\geq \sum_{q \in A_n} H_\xi(\alpha_n \mid T^{a, e_G}(q)) \mathbb{P}^\xi(T^{a, e_G}(q)) \\ &\geq cnH(\nu) \sum_{q \in A_n} \mathbb{P}^\xi(T^{a, e_G}(q)) \\ &= cnH(\nu) \mathbb{P}^\xi\left(\bigcup_{q \in A_n} T^{a, e_G}(q)\right) \\ &\geq pcnH(\nu). \end{aligned}$$

(5) We conclude that for every $n \geq 1$ we have

$$\frac{H_\xi(\alpha_n)}{n} \geq pcH(\nu),$$

and hence $h^\xi(\mu) > pcH(\nu) > 0$.

□

3. STRATEGY FOR THE PROOF OF THEOREM 1.1

- (1) Consider a finitely supported non-degenerate probability measure μ on T , and denote by (S^1, ν) the associated μ -boundary given by the circle endowed with the corresponding stationary measure.
- (2) We are going to prove the following statement.

Proposition 3.1. *For λ -almost every $\xi \in S^1$ we have $h^\xi(\mu) > 0$.*

- (3) We just need to prove that for λ -almost every $\xi \in S^1$ there is $C > 0$ such that for every n sufficiently large, we have $H_\xi(w_n) \geq Cn$.
- (4) Note that $h^\xi(\mu^{*N}) = Nh^\xi(\mu)$. Hence, since μ is non-degenerate, we may assume that the support of μ contains:
 - The identity element,
 - a non-trivial element supported on $[1/4, 3/4]$,
 - a non-trivial element supported on $[1/2, 1]$,
 - a non-trivial element supported on $[3/4, 1] \cup [0, 1/4]$, and
 - a non-trivial element supported in $[0, 1/2]$.
- (5) Let us consider boundary points $\xi \in [3/8, 5/8]$. Let a be the identity element, and b be the non-trivial element supported on $[3/4, 1] \cup [0, 1/4]$.
- (6) Let us consider trajectories of the μ -random walk for which ξ is the boundary point of S^1 .

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Email address, Martín Gilabert: gilabert@math.univ-lyon1.fr, martingilabertviolet@gmail.com

INSTITUT CAMILLE JORDAN, UNIVERSITÉ CLAUDE BERNARD LYON 1, 43 BOULEVARD DU 11 NOVEMBRE 1918, 69622 VILLEURBANNE, FRANCE.

Email address, Eduardo Silva: eduardo.silva@ens.fr, edosilvamuller@gmail.com

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DÉPARTEMENT DE MATHÉMATIQUES ET APPLICATIONS, ÉCOLE NORMALE SUPÉRIEURE, PSL RE-
SEARCH UNIVERSITY, 45 RUE D'ULM, 75005, PARIS, FRANCE.